

- 4 (a) (i) change of shape / size / length / dimension C1
 when (deforming) force is removed, returns to original shape / size A1 [2]
- (ii) $L = ke$ B1 [1]
- (b) $2e$ B1
 $\frac{1}{2}k$...(*allow e.c.f. from extension*) B1
- $\frac{1}{2}e$ and $2k$ B1
- $\frac{3}{2}e$...(*allow e.c.f. from extension in part 2*) B1
 $\frac{2}{3}k$...(*allow e.c.f. from extension*) B1 [5]

- 5 (a) either phase difference is π rad / 180°